INORGANIC CUMULATIVE EXAMINATION

January 19, 1991

Isotopic substitution can frequently be very informative in assigning solution structures or developing mechanisms for inorganic compounds. Answer all three parts of this exam which addresses various aspects of the preparation and utilization of isotopic labelling in inorganic chemistry.

PART I: Synthesis:

Suggest reasonable syntheses for 2 of the following compounds. Assume that any <u>common</u> reagent can be obtained in an isotopically enriched form as needed.

$^{33}SSO_3^{2-}$

<u>trans</u>-$Pt(^{15}NH_3)_2(^{14}NH_3)_2^{2+}$

$Co(NH_3)_5{}^{18}ON^{16}O^{+2}$

PART II: Spectroscopy:

Describe three different spectroscopic techniques that can use isotopic substitution to obtain useful molecular information. Briefly, note the nature of the experimental quantity obtained. Comment on any special requirements placed upon the isotope.

PART III:

Give one example of the use of isotopic labels in mechanistic or structural studies of inorganic compounds.